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DHA reduces hypothalamic inflammation and improves central leptin signaling in mice

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Abstract:

Aims: Anti-obesity effects and improved leptin sensitivity from n-3 polyunsaturated fatty acids (n-3 PUFAs) have been reported in diet-induced obese animals. This study sought to determine the beneficial central effects and mechanism of docosahexaenoic acid (DHA, 22:6 n-3) in high-fat (HF) diet fed mice.

Main methods: Male C57BL/6J mice were given HF diet with or without intracerebroventricular (icv) injection of docosahexaenoic acid (DHA, 22:6 n-3) for two days. Central leptin sensitivity, hypothalamic inflammation, leptin signaling molecules and tyrosine hydroxylase (TH) were examined by central leptin sensitivity test and Western blot. Furthermore, the expression of hepatic genes involved in lipid metabolism was examined by RT-PCR.

Key findings: We found that icv administration of DHA not only reduced energy intake and body weight gain but also corrected the HF diet-induced hypothalamic inflammation. DHA decreased leptin signaling inhibitor SOCS3 and improved the leptin JAK2-Akt signaling pathways in the hypothalamus. Furthermore, icv administration of DHA improved the effects of leptin in the regulation of mRNA expression of enzymes related to lipogenesis, fatty acid β -oxidation, and cholesterol synthesis in the liver. DHA increased leptin-induced activation of TH in the hypothalamus.

Significance: Therefore, increasing central DHA concentration may prevent the deficit of hypothalamic regulation, which is associated with disorders of energy homeostasis in the liver as a result of a high-fat diet.

Keywords: DHA, hypothalamic inflammation, leptin resistance, obesity

Abbreviations:

DHA, docosahexaenoic acid; TNF- α , Tumor Necrosis Factor α ; IL-1 β , interleukin 1 β ; IL-6, interleukin 6; JAK2, Janus kinase 2; SOCS3, suppressor of cytokine signaling 3; HF, high-fat diet; JNK, Jun N-terminal Kinase; FAS, fatty acid synthase; SCD1, stearoyl-coenzyme A desaturase 1; SFA, saturated fatty acids; PUFAs, polyunsaturated fatty acids; TH, tyrosine hydroxylase; MBH, mediobasal hypothalamus; PVN, paraventricular nucleus

1. Introduction

Leptin, secreted by adipocytes, acts centrally, particularly in the hypothalamus, to promote negative energy balance, reducing food intake and body weight (Zhang et al. , 1994). Hypothalamic leptin resistance plays a critical role in the development of obesity and its related metabolic disorders, such as hyperlipidemia (Bjorbaek and Kahn, 2004, Posey et al. , 2009). Improving central leptin sensitivity breaks the vicious cycle of energy balance dysregulation, as evidenced in obesity. The sensitivity of hypothalamic leptin to various fatty acids plays an essential role in the complex network of leptin signals controlling energy metabolism. High-fat (HF) diet with saturated fatty acids has been shown to decrease leptin sensitivity and inhibit the anorexigenic effect of leptin in rodents (El-Haschimi et al. , 2000, Kleinridders et al. , 2009). The n-3 polyunsaturated fatty acids (n-3 PUFAs) have been shown to exert some beneficial effects against obesity and diabetes (Cintra et al. , 2012, Pimentel et al. , 2012, Rossmeisl et al. , 2009). However, the role and mechanism of docosahexaenoic acid (DHA, 22:6 n-3), one of the most abundant n-3 PUFAs in body weight and hyperlipidemia management and modulation of the hypothalamic leptin signaling remain largely undiscovered.

Leptin signaling pathway includes the Janus kinase 2 (JAK2) - the phosphatidylinositol 3-kinase (PI3K) - Akt signaling in the central nervous system (Buettner et al. , 2006). In diet-induced obese rodents, leptin-induced pJAK2 and pAkt activation in the hypothalamus are attenuated (Levin et al. , 2004, Munzberg et al. , 2004). This is accompanied by an increased mRNA level of suppressor of cytokine signaling 3 (SOCS3), a negative regulator of the leptin signaling, which induces leptin resistance (Munzberg et al., 2004). In addition, it has been evident that the leptin PI3K signaling pathway is impaired during the development of diet-induced obesity in FVB/N mice (Metlakunta et al. , 2008). Therefore, it seems that the defective regulation of the above leptin pathways in the

hypothalamus plays a role in the pathogenesis of leptin resistance and obesity.

Leptin binds to long-form LepRb on neurons in several regions in the hypothalamus, including the mediobasal hypothalamus (MBH) and paraventricular hypothalamic nuclei (PVN), to regulate food intake and energy homeostasis (Metlakunta et al., 2008). HF diet-induced central leptin insensitivity and impaired hypothalamic leptin signaling are associated with the hypothalamic inflammation (Carvalho et al. , 2003, Posey et al., 2009). While n-3 PUFAs have shown anti-inflammatory effects by reducing the intracellular inflammatory mediators (Cintra et al., 2012, Morin et al. , 2011). For instance, dietary fish oil (n-3 PUFAs) enriched in HF diet feeding for two months was found to reduce the hypothalamic levels of TNF- α and IL-6, and increase the level of the anti-inflammatory cytokine IL-10 in male Wistar rats (Pimentel et al. , 2013). However, it is unknown if the anti-inflammation effects in the hypothalamus are due to the direct central effects of n-3 PUFAs.

It is reported that hypothalamic leptin mediates peripheral lipid metabolism by regulating the genes of key enzymes involved in its metabolism in the liver (Prieur et al. , 2008). An intracerebroventricular (icv) injection of leptin inhibits hepatic lipogenesis, promotes fatty acid oxidation, and cholesterol transportation (Minokoshi et al. , 2002). Prieur and colleagues have demonstrated that icv administration of leptin not only suppresses lipogenesis by decreasing fatty acid synthase (*FAS*), and stearoyl-coenzyme A desaturase 1 (*SCD1*) mRNA expression, but also activates β -oxidation by increasing acetyl-CoA acetyltransferase 1 (*ACAT1*) mRNA expression in the liver of ob/ob mice (Prieur et al., 2008). Finally, it has been suggested that leptin regulates peripheral energy metabolism through its action on autonomic nerves, particularly via the sympathetic nerve system, which transmits leptin signals to the liver and other tissues associated with energy metabolism (Ohno et al. , 2013, Warne et al. , 2011).

DHA in fish oils and fatty fish has been shown to have a variety of beneficial effects, such as improving dyslipidemia, reducing inflammation, and improving body weight management (Acharya and Talahalli, 2019, Sun et al. , 2018, Tocher et al. , 2019). Most research of DHA was epidemiological studies or dietary intervention studies. In this study, we examined the central effects of icv administration of DHA on leptin sensitivity, leptin signaling and inflammation in the hypothalamic brain region (MBH and PVN) of mice fed an acute HF diet; and further examined the effects of icv DHA and leptin on the expression of hepatic genes involved in lipid metabolism and the protein content of tyrosine hydroxylase (TH) in the hypothalamus for sympathetic control.

2. Materials and methods

2.1. Animals

Seventy-two male C57BL/6J mice (10 weeks old, body weight: 25.35 ± 1.83 g) were obtained from the Animal Resource Centre (Perth, WA, Australia) and housed in environmentally controlled conditions (temperature 22 °C, 12 hour light/dark cycle). All experimental procedures were approved by the Animal Ethics Committee, University of Wollongong, Australia, and comply with the *Australian Code of Practice for the Care and Use of Animals for Scientific Purposes*.

2.2. Cannula implantation

After 1 week of acclimatization, all mice were anesthetized by isoflurane inhalation and placed in a stereotactic device. A cannula was implanted into the right lateral brain ventricle (0.25 mm posterior and 1.0 mm lateral relative to Bregma and 2.5 mm below the surface of the skull) as described in our previous study (Wu et al. , 2014). The accuracy of cannula implantation into the lateral ventricle was confirmed by examining the needle track in the brain sections of each animal.

2.3. Animal experiment and energy homeostasis test

Seven days after the cannula implantation, the mice were randomized into three groups (n=24): 1) lab chow (LC) group with icv vehicle; 2) HF group with icv vehicle; 3) HF + icv DHA group. LC group were fed LC diet (5% kcal fat, 3.21 kcal/g) (Vella Stock feeds, Doonside, NSW, Australia), and the other two groups were fed HF diet (60% kcal fat, 4.58 kcal/g) for 2 days. The food composition data of HF diet are as follows (% g): total fat (soy oil, 3%, and lard, 33%): 36%, carbohydrate: 31%, protein: 20%, others (bran, minerals, and vitamins): 13% (Thaler et al. , 2012). The mice in the HF groups were treated with icv injection of DHA (1.5 nmol/2 μ l) or vehicle (2 μ l) for two days (twice/day— 09:00 and 15:00) (Abizaid and Horvath, 2008). The mice in LC group were treated with vehicle injection on the first two days when the other mice received DHA icv injection. Energy intake and body weight were measured after the icv DHA injection for 48 hours. As described previously (Ross et al. , 2010), DHA (D2534, Sigma-Aldrich) was dissolved in 96% ethanol, dried using nitrogen gas, dissolved in 40% hydroxypropyl- β -cyclodextrin (HPB) (H107, Sigma-Aldrich), and stored as stock solutions (100 nmol) at -20 °C. The working solution was made by diluting the stock solution with saline and contained 1.5 nmol DHA for every injection. 40% HPB without DHA was diluted in saline as vehicle control. The central leptin sensitivity test and tissue collection were performed detailed below. The experiment procedure was shown in Supplementary Fig. 1.

2.4. Central leptin sensitivity test

Two days after the HF diet with or without DHA icv injection, a central leptin sensitivity test was performed as described previously (Yu et al. , 2013). The mice were randomly divided into two subgroups (n = 12), received an icv injection of either leptin (0.5 μ g in 2 μ l) or 2 μ l saline. Energy intake was then measured 48 h after the icv leptin or vehicle injection.

2.5. Tissue collection

After central leptin sensitivity test, mice receiving either an icv injection of leptin (0.5 μ g in 2 μ l saline) or saline (2 μ l). Thirty minutes later, mice were sacrificed by CO₂ asphyxiation to examine the hypothalamus and liver tissue in response to leptin (Cheng et al. , 2015, Prieur et al., 2008). The mice were sacrificed thirty minutes after icv injection of leptin, as leptin signaling in the hypothalamus was significantly response to leptin as early as thirty minutes (Metlakunta et al. , 2011). The brains and livers were immediately collected, snaps frozen in liquid nitrogen, and stored at -80°C for further processing and analysis. At a temperature of -18°C, 500 μ m frozen brain sections were cut using a cryostat from Bregma-0.58 mm to -2.72 mm according to a standard mouse brain atlas (Paxinos G, 2002). The mediobasal hypothalamus (MBH) and the paraventricular nucleus (PVN) of the hypothalamus were dissected from frozen coronal sections using a Stoelting Brain Punch (#57401, 0.5 mm diameter, Wood Dale, Stoelting Co, USA) based on previously described coordinates (Yu et al., 2013). The brain tissues from MBH or PVN were punched for protein extraction (n=6) and RNA extractions (n=6).

2.6. Western Blot analysis

Western blotting was performed on protein extracts from frozen tissue as described in our previous study (Wu et al., 2014). The expression of specific proteins was determined using the following antibodies: pI κ B α Ser32 (sc-8404), and pJAK2 Tyr1007/Tyr1008 (sc-21870) from Santa Cruz Biotechnology (Santa Cruz, CA, USA), and SOCS3 (#2932), pAkt Ser473 (#9271), and pJNK Thr183/Tyr185 (#9251) from Cell Signalling Technology (Beverly, MA, USA), and TH (Anti-Tyrosine Hydroxylase (AB9983) from EMD Millipore (USA). I κ B α (AF1282) from Beyotime Technology (Shanghai, China). The concentration of antibodies was given in Table S1 for details. Bands corresponding to the proteins of interest were scanned, and band density was

analyzed using the Quantity One automatic imaging analysis system (Bio-Rad Laboratories, Hercules, CA, USA). All quantitative analyses were normalised to β -actin, based on our previous study (Yu et al., 2013). Due to the small amount of tissue in the MBH and PVN of the hypothalamus, we used a previously-described modified multi-strip Western Blot, which allows the detection of multiple proteins with a smaller sample size than the standard Western Blot (Yu et al., 2013).

2.7. RNA isolation and RT-PCR

Total RNA from the hypothalamus and liver was extracted using the Aurum total RNA mini kit (Bio-Rad Laboratories, Hercules, CA, USA) and reverse-transcribed to first-strand complementary DNA using the high-capacity cDNA reverse transcription kit (AB Applied Biosystems, CA, USA) according to the manufacturer's instructions. Quantitative real-time PCR (qPCR) was performed to analyze the resultant cDNA using 2X SYBR green I master mix on a Lightcycler 480 Real-time PCR System (F. Hoffmann-La Roche Ltd, Switzerland). Reactions were done in duplicate for each biological sample. The relative mRNA expression level for each gene was calculated by the $2^{(-DDCt)}$ method and normalized to the housekeeping gene β -actin or GAPDH. A list of gene-specific primers is available in Table S2.

2.8. Statistical analysis

Data were analyzed using the statistical package SPSS 19.0 (SPSS, Chicago, IL, USA). One-way analysis of variance (ANOVA) and the post hoc Tukey–Kramer honestly significant difference (HSD) test were used to analyze central leptin sensitivity, hypothalamic leptin signaling molecules, and the mRNA expression of inflammatory cytokines. $p < 0.05$ was regarded as statistically significant. Values are expressed as mean $\pm$ SEM.

3. Results

3.1. DHA reduced energy intake and body weight gain in HF diet fed mice

The HF diet significantly increased energy intake (+44.85%, $p<0.001$, Fig. 1A) and body weight gain (+149.90%, $p<0.001$, Fig. 1B) in 48 hours in comparison to the lab chow group. To examine the direct effect of central administration of n-3 PUFAs on the regulation of energy intake and body weight gain, the icv injection of DHA was given in the HF group. The icv injection of DHA decreased energy intake (-15.22%, $p<0.01$, Fig. 1A) and body weight gain (-51.85%, $p<0.05$, Fig. 1B) compared to the icv injection of vehicle in the HF group. These results indicate that the central administration of DHA exerts an anorexigenic effect in the HF diet-fed mice.

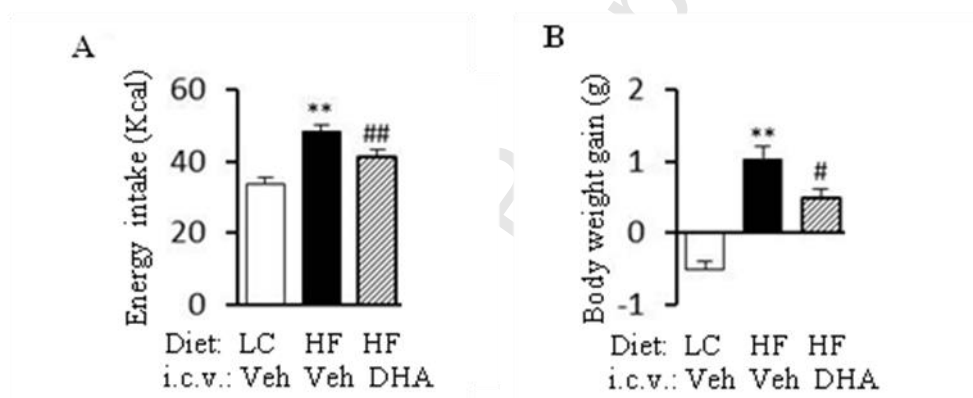


Figure 1. The effect of icv DHA on energy intake and body weight gain.

Energy intake (A) and body weight gain (B) for 48 hours were significantly increased in the HF-diet group, and inhibitory effects were induced by the icv injection of DHA. $n=12/\text{group}$. ** $p<0.01$ vs LC+Veh; # $p<0.05$, ## $p<0.01$ vs HF+Veh.

3.2. DHA reduced hypothalamic inflammation in HF diet fed mice

Previously, it is reported that hypothalamic inflammation occurs very rapidly in response to HF diet (Thaler et al., 2012). In both the MBH and the PVN of the hypothalamus, HF diet increased the mRNA expression of proinflammatory cytokines, including *TNF- α* (both $p<0.05$, Fig. 2A & 2E),

IL-1 β (MBH: $p<0.01$, Fig. 2A; PVN: $p<0.05$, Fig. 2E), and *IL-6* (both $p<0.05$, Fig. 2A & 2E). To determine the central effects of DHA on hypothalamic inflammation, the expression of proinflammatory cytokines and signaling molecules was measured in response to icv injection of DHA in the HF diet-fed mice. Administration of DHA centrally inhibited the mRNA expression of *TNF- α* (PVN: $p<0.05$, Fig. 2E), *IL-1 β* (MBH: $p<0.05$, Fig. 2A; PVN: $p=0.091$, Fig. 2E), *IL-6* (both $p<0.05$, Fig. 2A & 2E). Furthermore, the protein levels of two important inflammatory signaling molecules pI κ B α and pJNK were upregulated in the MBH and PVN of HF mice (both $p<0.05$, Fig. 2B & 2F & 2C & 2G). DHA icv injection significantly inhibited HF diet-induced pI κ B α /I κ B α and pJNK/JNK level in the MBH (Fig. 2B & 2C), but not in the PVN (Fig. 2F & 2G). These data demonstrate that the central administration of DHA exerts a potent anti-inflammatory response in the hypothalamus, especially in the MBH of HF mice.

Activation of I κ B α signaling pathway in the hypothalamus increases SOCS3 (Tocher et al., 2019), which is an inhibitor of leptin signaling molecules, JAK2 and Akt, and it has been suggested to mediate central leptin resistance in obesity (Gallardo et al. , 2007). In the current study, the level of SOCS3 was significantly elevated in the MBH and PVN of the HF mice (both $p<0.05$, Fig. 2D & 2H). This effect was inhibited by central DHA injection in the MBH ($p<0.05$, Fig. 2D).

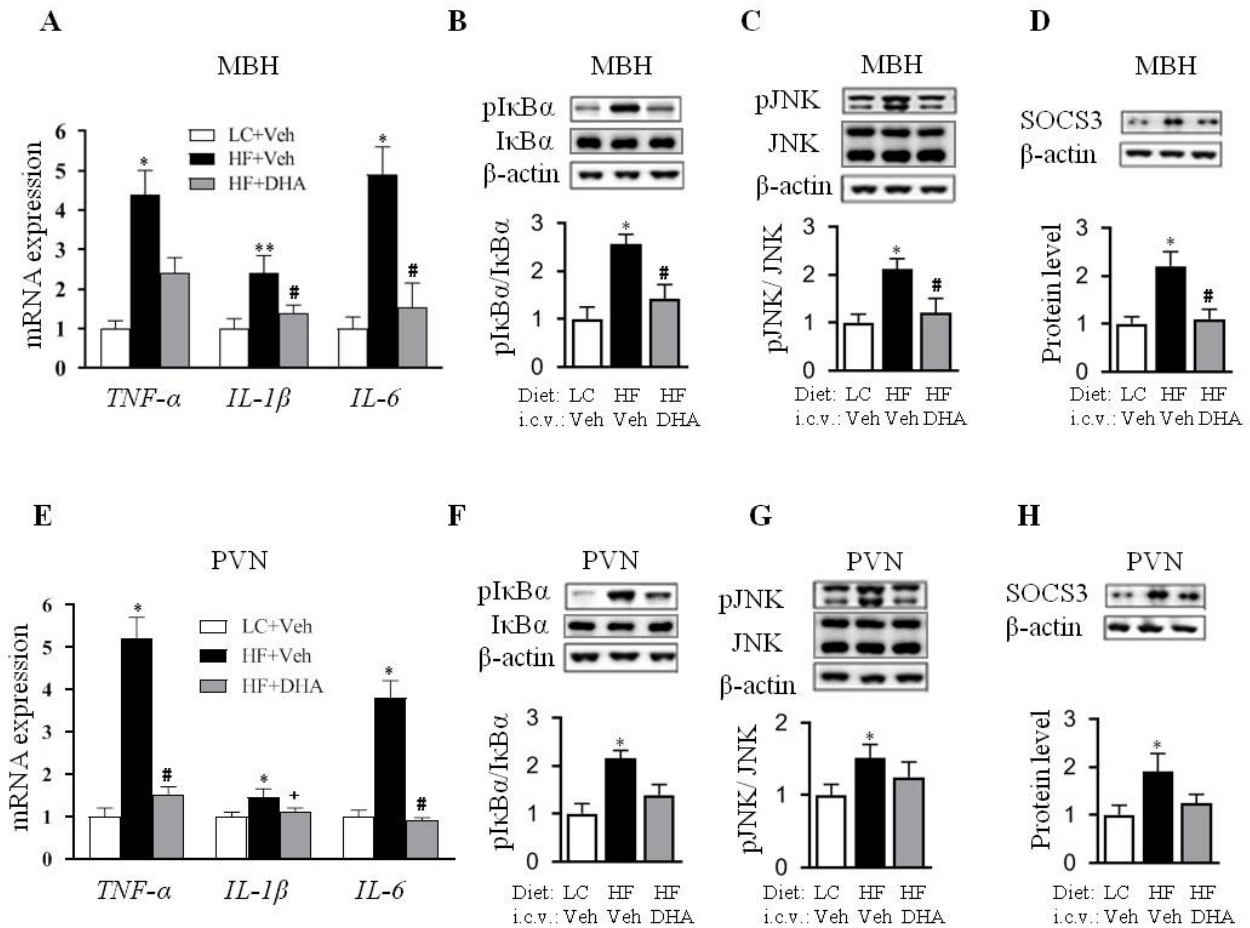


Figure 2. The effect of icv DHA on the inflammatory response in the hypothalamus.

The mRNA expression of *TNF-α*, *IL-1β*, and *IL-6* in the MBH (A) and PVN (E) was detected by RT-PCR. The protein levels of pIkBα (B and F), pJNK (C and G), and SOCS3 (D and H) in the MBH and PVN were detected by Western Blot after the icv injection of DHA and vehicle in the HF mice. n=6/group. *p<0.05, **p<0.01 vs LC+Veh; #p<0.05, +0.05<p<0.1 vs HF+Veh.

3.3. DHA improves leptin signaling in the hypothalamus of HF diet fed mice

Since DHA decreased leptin signaling inhibitor SOCS3, we next examined hypothalamic leptin signaling, JAK2, and Akt, important for negative energy homeostasis (Myers, 2004, Plum et al. , 2006). In the lab chow feed group, the level of pJAK2 and pAkt was significantly increased following the leptin icv injection (Supplementary Fig. 2). However, in the HF group, the icv

injection of leptin failed to activate above signaling molecules in the MBH and PVN, suggesting HF diet impaired central leptin signaling (Fig. 3 A-D). With icv injection of DHA, leptin significantly increased pJAK2 (MBH: $p<0.05$, Fig. 3A) and pAkt (MBH: $p<0.01$, Fig. 3B; PVN: $p<0.05$, Fig. 3D) compared with saline vehicle injection in HF diet group. DHA icv injection significantly increased pJAK2 in the PVN ($p<0.05$, Fig. 3C). These data suggest that the central administration of DHA improves leptin signaling JAK2 and Akt pathways in the hypothalamus. Furthermore, the icv injection of DHA significantly improved central leptin sensitivity in HF fed mice, evidenced by in the central leptin sensitivity test, energy intake for 48 hours was significantly decreased after icv injection of leptin compared with saline injection in the LC and HF+DHA mice, but not in the HF mice (Supplementary Fig. 3).

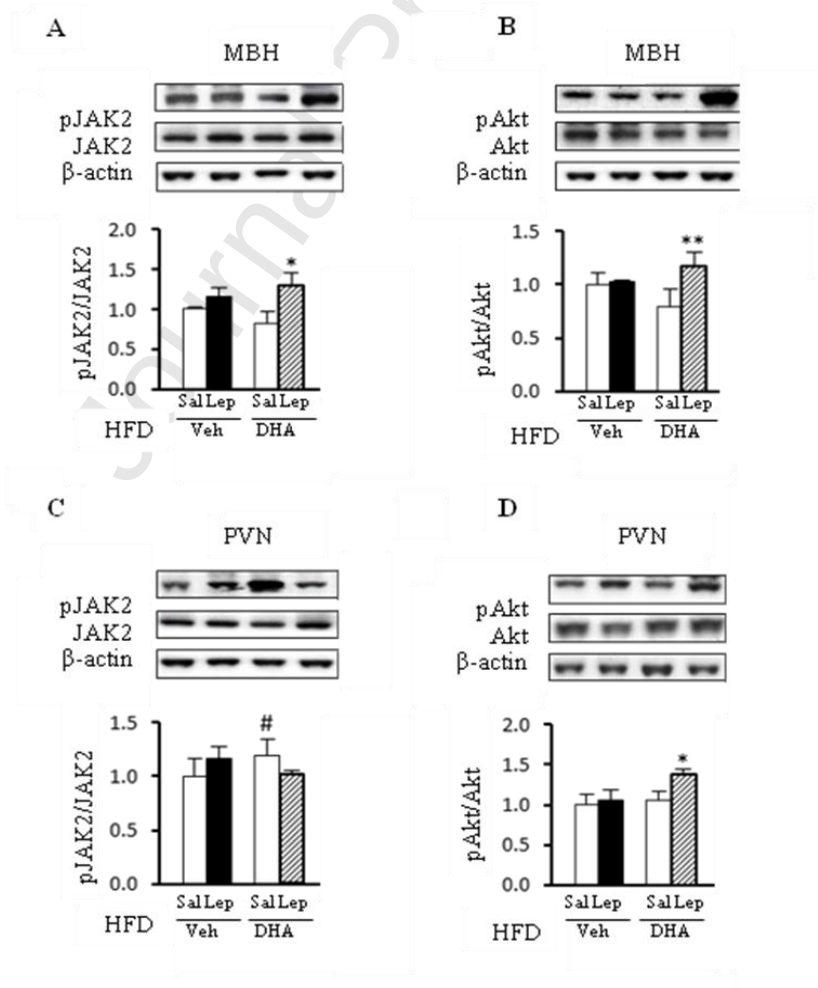


Figure 3. icv DHA improves leptin JAK2 and Akt signaling in the hypothalamus.

Phospho-JAK2 (pJAK2) (A and C) and phospho-Akt (pAkt) (B and D) in the MBH and PVN were detected by Western Blot after an icv injection of leptin or saline followed by the icv injection of DHA and vehicle in the HF diet-fed mice. The membranes were also reprobbed with antibodies against the total non-phosphorylated form of JAK2 (A and C) and Akt (B and D) in the MBH and PVN. n=6/group. * $p<0.05$, ** $p<0.01$ vs saline injection, # $p<0.05$ vs HF+Veh.

3.4. DHA improves the central leptin in the regulation of the expression of genes involved in hepatic lipid metabolism in HF diet-fed mice

Next, we used qRT-PCR to examine the mRNA expression of enzymes involved in hepatic lipogenesis such as *FAS*, *SCD1*, and *SREBP-1c* to determine the effects of icv DHA in response to central leptin. We have reported that central leptin significantly decreased the level of genes for lipogenesis, *FAS*, and *SCD1* in the liver (Cheng et al., 2015). Here, in the HF group, leptin central administration did not alter the gene expression of *FAS*, *SCD1*, and *SREBP-1c* compared with the vehicle injection (Fig. 4A-C), suggesting HF diet feeding may resist the action of leptin inhibiting lipogenesis. Interestingly, with DHA central administration, *FAS*, *SCD1*, and *SREBP-1c* mRNA expression were decreased in response to the leptin injection (all $p<0.05$). Also, the *SCD-1* mRNA expression level of DHA + Saline groups increased compared with HF + Vehicle groups with a statistical trend ($p<0.01$, Fig 4B). The mRNA expression of *FAS* and *SREBP-1* was not significant but tended to increase compared with HF + Vehicle groups.

The *ACAT1* is an important enzyme involved in mitochondrial β -oxidation (Fournier et al. , 1994). Peroxisome proliferator-activated receptor alpha (*PPAR α*) is a key regulator in the regulation of

fatty acid β -oxidation (Dreyer et al. , 1993). Here, in the HF group, leptin icv injection could not increase the levels of these two enzymes. In contrast, after the DHA pre-treatment, the mRNA expression of *ACAT1* and *PPAR α* was profoundly increased by leptin compared with the vehicle treatment (both $p < 0.05$, Fig. 4D&4E). These results suggest that DHA ameliorates the central leptin modulation of β -oxidation in the liver. Furthermore, 3-hydroxy-3-methylglutaryl-coenzyme reductase (*HMG-CoA reductase*) is the key enzyme involved in the de novo synthesis of cholesterol. In the DHA pre-treatment group, leptin icv injection reduced *HMG-CoA* expression in liver compared with vehicle injection ($p < 0.01$, Fig. 4F), but not in the HF group without DHA pre-treatment.

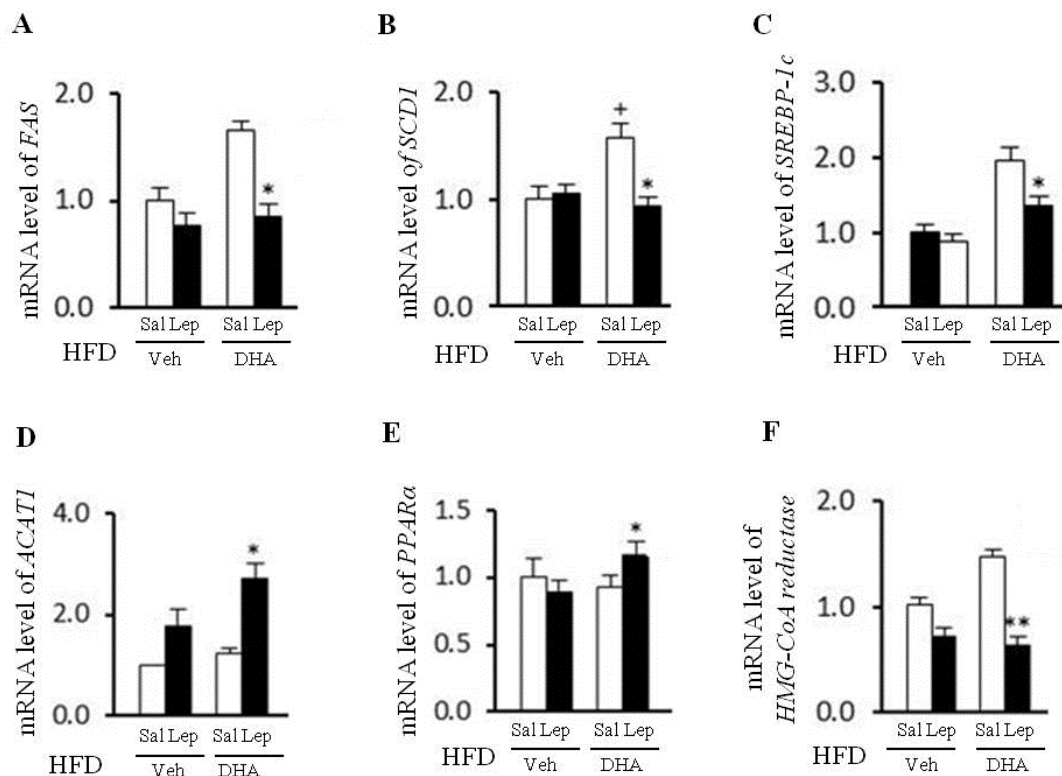


Figure 4. Effects of icv DHA on mRNA expression of genes involved in hepatic lipogenesis, fatty acid β -oxidation, and cholesterol metabolism.

The mRNA levels of *FAS* (A), *SCD1*(B), *SREBP-1c*(C), *ACAT1* (D), *PPAR α* (E) and *HMG-CoA reductase* (F) in the liver were measured by quantitative real-time PCR in HF diet-fed mice treated with an icv injection of leptin or saline after the icv injection of DHA or vehicle for 2 days. n=6/group. * $p<0.05$, ** $p<0.01$ vs saline injection, +0.05< $p<0.1$ vs HF+Veh.

3.5. DHA enhances hypothalamic TH expression

To determine whether DHA regulates leptin-mediated hepatic metabolism through the sympathetic nervous system, the protein level of hypothalamic TH was examined. We found that leptin could significantly increase TH level in the MBH of HF group with DHA pre-treatment ($p<0.05$, Fig. 5A), but not in the HF group without DHA pre-treatment, suggesting DHA improving leptin in the regulation of sympathetic nervous tone in the MBH. However, in the PVN, central leptin significantly increased TH level compared with the vehicle icv injection in the HF group ($p<0.05$, Fig. 5B), while with DHA pre-treatment, leptin icv injection did not increase TH level ($p<0.01$, Fig. 5B). These results suggest that DHA regulates leptin-mediated hepatic metabolism mainly through the sympathetic nervous system in the MBH of the hypothalamus.

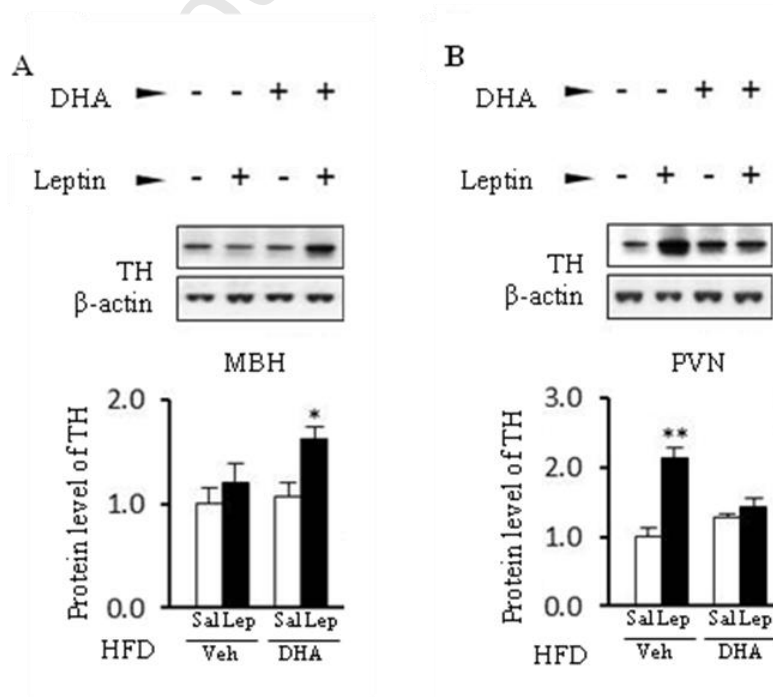


Figure 5. Effects of icv DHA on TH protein level in the hypothalamus.

The level of TH protein expression in the MBH (A) and PVN (B) of the hypothalamus was detected by Western Blot in HF diet-fed mice treated with an icv injection of leptin or saline after the icv injection of DHA or vehicle for 2 days. n=6/group. *p<0.05, **p<0.01 vs saline injection.

4. Discussion

It has been reported that n-3 PUFAs are associated with the prevention of HF diet-induced obesity *in vivo* and *in vitro* studies (Mori et al. , 2004, Reseland et al. , 2001, Storlien et al. , 1987, Ukropec et al. , 2003). However, the potential effects of central n-3 PUFAs on energy homeostasis are largely unrevealed. In the present study, acute HF diet feeding induced a significant increase of energy intake and body weight gain. Importantly, the icv administration of DHA decreased energy intake and body weight gain induced by HF diet, which is consistent with previous study by Schwinkendorf's group who have shown that the central administration of DHA reduced food intake and body mass in male Sprague-Dawley rats with lab chow diet (Schwinkendorf et al. , 2011). Here, we further demonstrated the central effects of DHA in inhibiting energy intake not only in the control diet but also in obesogenic diet consumption.

Dietary fatty acids could be attractive candidates for either promoting or controlling the hypothalamic inflammation, which is important in the initialization and progression of diet-induced obesity. It has been reported that both HF diet and the central administration of saturated fatty acid (SFA) can induce an inflammatory response in the hypothalamus (Kleinridders et al., 2009, Milanski et al. , 2009). Here, we found the HF diet for two days induced a proinflammatory response in the hypothalamus (as demonstrated by the increased mRNA expression of *TNF- α* , *IL-1 β* , and *IL-6*), which is consistent with the previous study of Thaler et al (Thaler et al., 2012). Importantly, the present study showed that the central administration of DHA reduced the

inflammatory cytokines and downstream inflammatory signaling molecules, including *TNF- α* , *IL-1 β* , *IL-6*, pI κ B α , and pJNK in the hypothalamus. The activation of the hypothalamic inflammatory molecule I κ B α has shown to promote central leptin resistance by increasing SOCS3 mRNA expression in mice (Tocher et al., 2019). SOCS3 is a mediator cross-linking the inflammation and leptin resistance. Therefore, hypothalamic inflammation in activation of SOCS3 may contribute to the impairment of central leptin signaling and the disinhibition of energy intake in HF diet consumption, while DHA's anti-inflammatory effects in the hypothalamus may contribute to its improvement in leptin sensitivity.

Previous research has shown that SOCS3 inhibits JAK2, the important downstream molecule of leptin signaling (Bjorbaek et al. , 1999, Bjorbaek and Kahn, 2004). Defective JAK2 signaling in the hypothalamus contributes to the development of HF diet-induced central leptin resistance and obesity (Bjornholm et al. , 2007, Ghilardi et al. , 1996, Munzberg et al., 2004). In the current study, we confirmed the hypothalamic JAK2 pathway of leptin signaling was impaired in the HF diet-fed mice. In the control group, leptin significantly increased pJAK2 in the MBH and PVN; however, leptin could not increase pJAK2 in the HF diet-fed mice suggesting leptin-JAK2 signaling was impaired. Therefore, the impaired leptin-pJAK2 signaling may contribute to increased food intake and body weight gain observed in the HF diet-fed group. Importantly, our results demonstrate that the central administration of DHA up-regulates pJAK2 expression at the specific nuclei in the hypothalamus, which may represent a potential route to improve leptin signaling. Our finding supports previous studies showing that the icv administration of n-3 PUFA (linolenic acid) improves hypothalamic leptin signal transduction via the JAK2 pathway (Cintra et al., 2012). In addition, it has been proven that heterozygous SOCS3-deficient mice have an increased sensitivity to the weight-reducing effects of leptin, and they are resistant to the development of diet-induced obesity

(Howard et al. , 2004). These findings suggest a critical role of SOCS3 in mediating hypothalamic leptin resistance via the JAK2 pathway. The present study demonstrated that the level of pJAK2 was specifically decreased in the hypothalamus of HF diet-fed animals, and also that DHA increased hypothalamic pJAK2 levels. Therefore, both previous reports and the present results indicate that DHA may exert its beneficial effect via activation of the level of pJAK2 through SOCS3 inhibition.

Another potential mechanism for the development of obesity is the impaired regulation of the leptin Akt signaling pathway in the hypothalamus (Metlakunta et al., 2008, Warne et al., 2011). Both dietary and central administration of n-3 PUFAs, such as fish oil, has shown to enhance the protein level of pAkt (Cintra et al., 2012, Pimentel et al., 2012). In line with previous studies, we demonstrated that the central administration of DHA promoted leptin Akt signaling in the hypothalamus. Taken together, our findings and previous reports may help to explain the potential mechanism of n-3 PUFAs on leptin Akt signaling for central leptin sensitivity improvement in prevention body weight gain induced by HF diet.

We found that JAK2 was not activated in the PVN when it was stimulated by leptin after DHA administration in the HF mice, but both JAK2 and Akt were activated in the BMH. Furthermore, TH level was increased in response to leptin only in the BMH, but not in the PVN with DHA administration. These results suggest leptin signaling was more sensitive in the BMH than in the PVN with DHA administration. In the arcuate nucleus (ARC) of BMH, LepRb-expressing first-order neurons sense to leptin. These neurons express neuropeptides controlling energy balance, such as agouti-related peptide (AgRP), neuropeptide Y (NPY) and proopiomelanocortin (POMC)-derived peptides. Both NPY/AgRP- and POMC-containing neurons project extensively to other key hypothalamic nuclei, including the PVN, dorsomedial hypothalamic nucleus and lateral

hypothalamic area, that in turn send projections to intrahypothalamic and extrahypothalamic sites to regulate energy balance (Sohn, 2015). The PVN is the most thoroughly characterized hypothalamic interface area between the endocrine, autonomic, and somatic motor systems, influencing feeding behavior and energy metabolism (Sawchenko, 1998, Sawchenko and Swanson, 1983, Watts, 2000). The thyrotropin-releasing hormone (TRH) and corticotrophin-releasing hormone (CRH) expressed in the PVN are involved in the control of energy balance (Fekete et al. , 2000, Fekete et al. , 2004, Richard and Baraboi, 2004). Interestingly, a subset of the PVN neurons also express LepRb and leptin activates the LepRb pathways in these neurons, suggesting that leptin is able to regulate a subset of the PVN neurons both directly and indirectly via the ARC first-order neurons (Ghamari-Langroudi et al. , 2011). Therefore, the differential response of leptin signaling, JAK2 in the PVN compared to that of BMH in the DHA group, may be due to second-order neurons in the PVN regulated by BMH, besides leptin's direct effects in the PVN. Furthermore, research shows that lesions in the ventromedial hypothalamus (VMH) of BMH uniformly produced obesity. However, lesions in the PVN produced obesity in only half of the animals (Tokunaga et al. , 1986), suggesting the characteristics of obesity produced by PVN lesions differ from those resulting from BMH damage.

Central administration of leptin has been demonstrated to improve hepatic lipid metabolism through upregulating fatty acid oxidation and down-regulating lipogenesis (Gallardo et al., 2007, Toyoshima et al. , 2005). The effects of dietary n-3 PUFAs on improving hepatic fatty acid β -oxidation have been reported (Dobrzyn and Dobrzyn, 2006, Neschen et al. , 2007, Rossmeisl et al., 2009). However, the central effects of n-3 PUFAs in regulating hepatic lipid metabolism are still unclear. In the present study, we showed that the expression of *FAS*, *SCD1*, and *SREBP-1c* genes were down-regulated by icv injection of DHA in response to leptin administration. This suggests

that DHA improves the effect of leptin on hepatic lipogenesis. In addition, the effect of leptin on fatty acid β -oxidation may be improved by icv injection of DHA, as evidenced by the up-regulation of *ACAT1* and *PPAR α* mRNA expression in the liver. Thus, our results showed for the first time that DHA in the brain might play a significant role in regulating central leptin's action on hepatic lipid metabolism, particularly lipogenesis and fatty acid β -oxidation. The suppressive effect of leptin on cholesterol synthesis via *HMG-CoA* reductase in obesity has already been evidenced (Prieur et al., 2008). In the present study, the central administration of DHA also decreased the mRNA expression of *HMG-CoA* reductase in the liver. Taken together, our findings suggest that DHA inhibits the gene expression of enzymes involved in hepatic lipogenesis and cholesterol synthesis and improves hepatic lipid oxidation. This may reduce lipid deposition in the liver, thereby alleviating the lipotoxicity and obesity induced by HF diet. In the present study, as only gene expression was examined, the protein levels and enzyme activities of these genes for lipid metabolism require further confirmation with DHA treatment.

It has been suggested that the sympathetic nervous system plays a key role in connecting central leptin signals from the hypothalamus to the liver and other peripheral tissues (Morton, 2007). For instance, it has been reported that leptin acts in the hypothalamus to suppress lipogenesis via activating the sympathetic nervous system in both liver and white adipose tissues (Buettner et al., 2008, Warne et al., 2011). The expression of TH (a rate-limiting enzyme for the synthesis of catecholamines) in the hypothalamus has been identified as a major candidate and indicator for the regulation of sympathetic outflow in normal or obese rodents (Li et al., 2009, Shi et al., 2013). In the present study, we showed for the first time that DHA increased leptin-induced activation of TH in the MBH of the hypothalamus, suggesting that DHA improves the function of leptin-activated sympathetic outflow. This finding may connect the effects of n-3 PUFAs in regulating hepatic lipid

metabolism with the hypothalamic sympathetic nervous system. However, further valid evidence is needed to prove this mechanism.

Previous studies have shown that n-3 PUFA intake exerts a beneficial effect on the hepatic lipid homeostasis (Guo et al. , 2017, Lu et al. , 2016). Oral administration of n-3 PUFAs (fish oil) significantly inhibits the up-regulation of the lipogenesis gene, *SREBP-1*, mRNA expression in the liver induced by a high-fat diet in mice (Bargut et al. , 2014). While in the present study, we found that the icv injection of DHA in trend increased the mRNA of *SCD-1*, *FAS*, *SREBP-1c*, lipogenesis genes in the liver of HF mice. The exact central mechanism of DHA in the up-regulation of these hepatic lipogenesis genes is not clear. The PVN is a vital brain region to regulate Hypothalamic-Pituitary-Adrenal (HPA) axis, which controls many functions of the liver through neuroendocrine forward signaling pathways as well as negative feedback mechanisms, in health and disease (Petrescu et al. , 2018). In the present study, we found that the icv injection of DHA decreased inflammation response in the hypothalamic PVN of HF diet fed mice. Our result is consistent with a fish oil supplement study in mice (Liu et al. , 2013), in which mRNA expression of *TNF- α* decreases in the hypothalamus. Furthermore, the n-3 PUFAs supplementation reduces the activation of the HPA axis evidenced by that fish oil supplementation induces the down-regulation of mRNA expressions of hypothalamic *CRH* and pituitary *POMC* as well as lower plasma adrenocorticotrophin and cortisol concentrations (Liu et al., 2013). Cortisol reaches the liver and binds to glucocorticoid receptors to regulate lipid metabolic pathways for body growth and energy production and storage (Petrescu et al., 2018). It is known that the activity of the HPA axis is low in periods of rest and relaxation so that the liver has a reduced rate of lipolysis directing the lipids for storage and lipogenesis (Petrescu et al., 2018). Therefore, icv DHA may improve the PVN inflammation induced by the HF diet and thereafter inhibit the HPA axis and promote hepatic

lipogenesis genes expression. However, a research shows that in primary hepatocytes, DHA downregulates the gene expression of lipogenic enzymes, *SREBP-1*, *SCD-1* and *FAS* via G-protein-coupled receptor 40 (On et al. , 2019). Overall these findings suggest that the central and peripheral effects of DHA in regulation of lipogenesis genes may be different, which requires further investigation.

5. Conclusion

In summary, the present study demonstrates that increased central DHA concentration exhibits beneficial effects on body weight and food intake control in high-fat diet feeding. Exposure of central DHA can lead to significant anti-inflammatory response and improve leptin JAK2 and Akt signaling pathways in the hypothalamus. Moreover, central leptin's action in regulating hepatic lipid metabolism can be improved by central DHA. Therefore, DHA may play an important central role in preventing the deficit of hypothalamic regulation of peripheral energy metabolism induced by a high-fat diet.

Data availability statement

The datasets generated for such study are available on request to the corresponding author.

Ethics Statement

All animal experiments used in this study are strictly adhered to the international standards for the care and use of laboratory animals and were approved by the Ethical Committee for the University of Wollongong.

Author Contributions

The authors wish to thank the following individuals for their contributions: Dr. Tracy Maddocks for her technical support in intracranial surgery. L.C.C. and W.H.Q. performed the research, analysed the data, and wrote the manuscript. T.H., H.L.S., and Z.K.Y. analysed the data and wrote

the manuscript. Y.H.Y and X.F.H. contributed to the research study designing and reviewing and editing the manuscript.

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Conflict of Interest Statement

The authors declare no conflict of interest.

Consent for publication

Not applicable.

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Conflict of Interest Statement

The authors declare no conflict of interest.

Author Contributions

The authors wish to thank the following individuals for their contributions: Dr. Tracy Maddocks for her technical support in intracranial surgery. L.C.C. and W.H.Q. performed the research, analysed the data, and wrote the manuscript. T.H., H.L.S. and Z.K.Y. analysed the data and wrote the manuscript. Y.H.Y and X.F.H. contributed to the research study designing, and reviewing and editing the manuscript.

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